

There are a total of five problems. You have to solve the first four. Problem 5 is optional.

DFA Set A

Problem 1 (CO1): DFA and Regular Languages (15 points)

We define the last two digits of your Student ID to be AB [e.g: If your Student ID is 2102895, then A = 9, B = 5]

Given, $\Sigma = \{A, B, \#\}$. Consider the following languages over Σ .

$$L_1 = \{w : w \text{ starts with A}\}$$

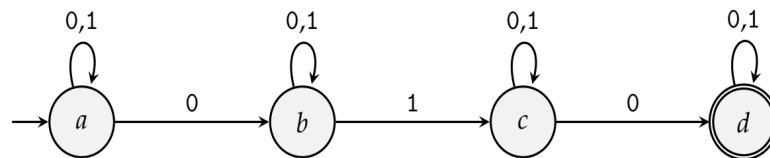
$$L_2 = \{w : w \text{ contains AB\# as a substring}\}$$

$$L_3 = L_1 \circ L_2$$

Now solve the following problems. For questions (a)-(f), you must use your specific Σ to answer.

- If $\Sigma = \{A, B, \#\}$, then **define** Σ according to your Student ID. (1 point)
- Give** the state diagram for a DFA that recognizes L_1 . (3 points)
- Give** the state diagram for a DFA that recognizes L_2 . (3 points)
- Find** all the four-letter strings in $L_1 \cap L_2$. (2 points)
- If you were to use the “cross product” construction shown in class to obtain a DFA for the language $L_1 \cap L_2$, how many states would it have? (1 point)
- Prove** L_3 is a regular language by giving the state diagram for a DFA or an NFA that recognizes L_3 . (2 points)

Now, let $\Sigma = \{0, 1\}$. Consider the following diagram of the NFA to answer the questions (g)-(h) defined for Σ .



- Choose** the language recognized by this NFA? (1 point)
 - $\{w : w \text{ has a length equal to or more than three.}\}$
 - $\{w : w = (010)^n, n \geq 0\}$
 - $\{w : w \text{ contains 010 as a subsequence}\}$
 - $\{w : w \text{ contains 010 as a substring}\}$
- Select** the paths that accepts 010110 in the given NFA? There can be more than one path that accepts the string. (2 points)
 - $a \rightarrow b \rightarrow b \rightarrow b \rightarrow b \rightarrow c \rightarrow d$
 - $a \rightarrow b \rightarrow c \rightarrow d \rightarrow d \rightarrow d \rightarrow d$
 - $a \rightarrow b \rightarrow b \rightarrow b \rightarrow b \rightarrow b \rightarrow b$
 - $a \rightarrow a \rightarrow b \rightarrow b \rightarrow c \rightarrow c \rightarrow d$
 - $a \rightarrow a \rightarrow a \rightarrow b \rightarrow c \rightarrow c \rightarrow d$

Problem 1 (CO1): DFA and Regular Languages (15 points)

We define the last two digits of your Student ID to be AB [e.g: If your Student ID is 2102895, then A = 9, B = 5]
Given, $\Sigma = \{A, B, \#\}$. Consider the following languages over Σ .

$$L_1 = \{w : w \text{ ends with } A\}$$

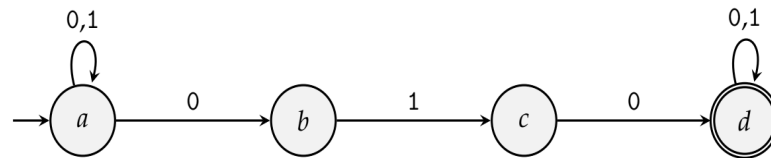
$$L_2 = \{w : w \text{ contains } B\#A \text{ as a substring}\}$$

$$L_3 = L_1 \circ L_2$$

Now solve the following problems. For questions (a)-(f), you must use your specific Σ to answer.

- (a) If $\Sigma = \{A, B, \#\}$, then **define** Σ according to your Student ID. (1 point)
- (b) **Give** the state diagram for a DFA that recognizes L_1 . (3 points)
- (c) **Give** the state diagram for a DFA that recognizes L_2 . (3 points)
- (d) **Find** all the four-letter strings in $L_1 \cap L_2$. (2 points)
- (e) If you were to use the “cross product” construction shown in class to obtain a DFA for the language $L_1 \cap L_2$, how many states would it have? (1 point)
- (f) **Prove** L_3 is a regular language by giving the state diagram for a DFA or an NFA that recognizes L_3 . (2 points)

Now, let $\Sigma = \{0, 1\}$. Consider the following diagram of the NFA to answer the questions (g)-(h) defined for Σ .



- (g) **Choose** the language recognized by this NFA? (1 point)
 - (i) $\{w : w \text{ has a length equal to or more than three.}\}$
 - (ii) $\{w : w = (010)^n, n \geq 0\}$
 - (iii) $\{w : w \text{ ends with } 010\}$
 - (iv) $\{w : w \text{ contains } 010 \text{ as a substring}\}$
- (h) **Select** the paths that accepts 010110 in the given NFA? There can be more than one path that accepts the string. (2 points)
 - (i) $a \rightarrow b \rightarrow b \rightarrow b \rightarrow b \rightarrow c \rightarrow d$
 - (ii) $a \rightarrow b \rightarrow c \rightarrow d \rightarrow d \rightarrow d \rightarrow d$
 - (iii) $a \rightarrow a \rightarrow a \rightarrow b \rightarrow c \rightarrow d \rightarrow d$
 - (iv) $a \rightarrow b \rightarrow b \rightarrow b \rightarrow b \rightarrow b \rightarrow b$
 - (v) $a \rightarrow a \rightarrow b \rightarrow b \rightarrow c \rightarrow c \rightarrow d$

Problem 1 (CO1): DFA and Regular Languages (15 points)

We define the last two digits of your Student ID to be AB [e.g: If your Student ID is 2102895, then A = 9, B = 5]

Given, $\Sigma = \{A, B, \#\}$. Consider the following languages over Σ .

$$L_1 = \{w : w \text{ starts with } \#\}$$

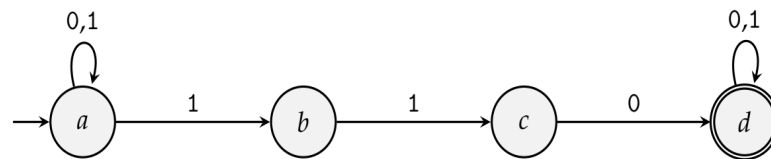
$$L_2 = \{w : w \text{ contains } \#BA \text{ as a substring}\}$$

$$L_3 = L_1 \circ L_2$$

Now solve the following problems. For questions (a)-(f), you must use your specific Σ to answer.

- If $\Sigma = \{A, B, \#\}$, then **define** Σ according to your Student ID. (1 point)
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- Find** all the four-letter strings in $L_1 \cap L_2$. (2 points)
- If you were to use the “cross product” construction shown in class to obtain a DFA for the language $L_1 \cap L_2$, how many states would it have? (1 point)
- Prove** L_3 is a regular language by giving the state diagram for a DFA or an NFA that recognizes L_3 . (2 points)

Now, let $\Sigma = \{0, 1\}$. Consider the following diagram of the NFA to answer the questions (g)-(h) defined for Σ .



- Choose** the language recognized by this NFA? (1 point)
 - $\{w : w \text{ has a length equal to or more than three.}\}$
 - $\{w : w = (110)^n, n \geq 0\}$
 - $\{w : w \text{ contains } 110 \text{ as a substring}\}$
 - $\{w : w \text{ contains } 010 \text{ as a substring}\}$
- Select** the paths that accepts 010110 in the given NFA? There can be more than one path that accepts the string. (2 point)
 - $a \rightarrow b \rightarrow b \rightarrow b \rightarrow b \rightarrow c \rightarrow d$
 - $a \rightarrow a \rightarrow a \rightarrow b \rightarrow c \rightarrow d \rightarrow d$
 - $a \rightarrow b \rightarrow c \rightarrow d \rightarrow d \rightarrow d \rightarrow d$
 - $a \rightarrow a \rightarrow a \rightarrow a \rightarrow b \rightarrow c \rightarrow d$
 - $a \rightarrow a \rightarrow b \rightarrow b \rightarrow c \rightarrow c \rightarrow d$

RE Set A

Problem 2 (CO1): Regular Expressions (15 points)

Let $\Sigma = \{0, 1\}$. **Give** regular expressions for each of the languages (a)-(f) over Σ .

- (a) $\{w : w \text{ contains } 11 \text{ or } 101 \text{ as a substring.}\}$ (2 points)
- (b) $\{w : w \text{ contains exactly four } 1\text{s.}\}$ (2 points)
- (c) $\{w : \text{The length of } w \text{ is two more than multiple of five.}\}$ (2 points)
- (d) $\{w : w \text{ consists of any combination of } 01 \text{ and } 110.\}$ (2 points)
- (e) $\{w : w \text{ doesn't end with } 01\}$ (2 points)
- (f) $\{w : \text{Number of } 01 \text{ substring is more than number of } 10 \text{ substrings in } w\}$ (2 points)
- (g) You write a regular expression $0(0+1)^*1^*0^*0$. Your friends write another regular expression $01^*0^*(0+1)^*0$. Are they the same? **Write** Yes or No only. (1 point)
- (h) You write a regular expression $(1+01)^*$. Your friends write another regular expression $1^*(011^*)^*$. Are they the same? **Give** justification for your answer. (2 points)

RE Set B

Problem 2 (CO1): Regular Expressions (15 points)

Let $\Sigma = \{0, 1\}$. **Give** regular expressions for each of the languages (a)-(f) over Σ .

- (a) $\{w : w \text{ starts with } 00 \text{ or } 010.\}$ (2 points)
- (b) $\{w : w \text{ contains at least three } 1\text{s.}\}$ (2 points)
- (c) $\{w : \text{The length of } w \text{ is three more than multiple of five.}\}$ (2 points)
- (d) $\{w : w \text{ consists of any combination of } 10 \text{ and } 001.\}$ (2 points)
- (e) $\{w : w \text{ doesn't end with } 11\}$ (2 points)
- (f) $\{w : \text{Number of } 01 \text{ substring is less than number of } 10 \text{ substrings in } w\}$ (2 points)
- (g) You write a regular expression $11^*(0+1)^*0^*1$. Your friends write another regular expression $10^*1^*(0+1)^*1$. Are they the same? **Write** Yes or No only. (1 point)
- (h) You write a regular expression $(0+10)^*$. Your friends write another regular expression $0^*(100^*)^*$. Are they the same? **Give** justification for your answer. (2 points)

RE Set C

Problem 2 (CO1): Regular Expressions (15 points)

Let $\Sigma = \{0, 1\}$. **Give** regular expressions for each of the languages (a)-(f) over Σ .

- (a) $\{w : w \text{ ends with } 001 \text{ or } 11.\}$ (2 points)
- (b) $\{w : w \text{ contains at most two } 1\text{s}.\}$ (2 points)
- (c) $\{w : \text{The length of } w \text{ is three more than multiple of four}.\}$ (2 points)
- (d) $\{w : w \text{ consists of any combination of } 11 \text{ and } 010.\}$ (2 points)
- (e) $\{w : w \text{ doesn't end with } 00\}$ (2 points)
- (f) $\{w : \text{Number of } 01 \text{ substring and } 10 \text{ substrings in } w \text{ is unequal}\}$ (2 points)
- (g) You write a regular expression $01^*(0+1)^*0^*1$. Your friends write another regular expression $00^*1^*(0+1)^*1$. Are they the same? **Write** Yes or No only. (1 point)
- (h) You write a regular expression $(1+01)^*$. Your friends write another regular expression $1^*(011^*)^*$. Are they the same? **Give** justification for your answer. (2 points)

Problem 3 (CO3): Converting Regular Expressions to NFAs (10 points)

Convert the following regular expression over $\Sigma = \{a, b, c\}$ into an equivalent NFA. Note that $R_1 + R_2$ is the same as $R_1 \cup R_2$.

$$(bc)^*(a + c) + (bc^* + a)^*b$$

RE to NFA Set A

Problem 3 (CO2): Converting Regular Expressions to NFAs (10 points)

Convert the following regular expression over $\Sigma = \{0, 1, 2\}$ into an equivalent NFA. Note that $R_1 + R_2$ is the same as $R_1 \cup R_2$.

$$(0^*1 + 2)^* + 0(12)^*0$$

RE to NFA Set B

Problem 3 (CO2): Converting Regular Expressions to NFAs (10 points)

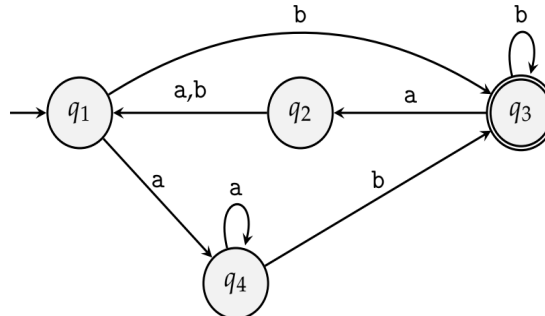
Convert the following regular expression over $\Sigma = \{0, 1, 2\}$ into an equivalent NFA. Note that $R_1 + R_2$ is the same as $R_1 \cup R_2$.

$$1^*0 + (0^*2 + (20)^*1)$$

DFA to RE Set A

Problem 4 (CO2): Converting Finite Automata to Regular Expressions (10 points)

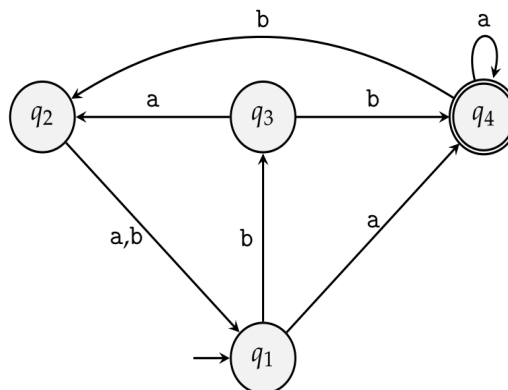
Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_3 , then q_2 , next q_4 , and finally q_1 . You must show work.



DFA to RE Set B

Problem 4 (CO2): Converting Finite Automata to Regular Expressions (10 points)

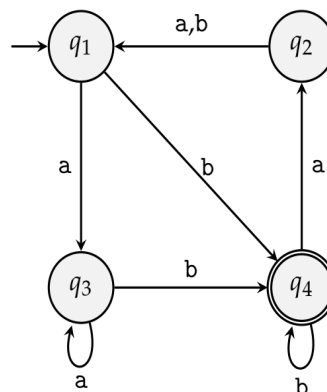
Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_4 , then q_2 , next q_3 , and finally q_1 . You must show work.



DFA to RE Set C

Problem 4 (CO2): Converting Finite Automata to Regular Expressions (10 points)

Convert the following DFA into an equivalent regular expression using the state elimination method. First eliminate q_4 , then q_2 , next q_3 , and finally q_1 . You must show work.



Problem 5 (Bonus): Even Odd (5 points)

Disclaimer: This is a bonus problem. Attempt it only after you are done with everything else. Even if you do not attempt it, you can get a perfect score. So, do not worry if you find it too hard!

Let $\Sigma = \{0, 1\}$.

$L = \{w \mid \text{0s in even position of } w \text{ are followed by odd numbers of 1s}\}$

Give a five state diagram for a DFA that recognizes L .

